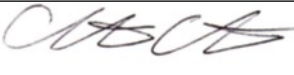


Guideline Title: Pediatric Sepsis or Meningitis
Guideline Number: GUID-3203-PED016

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 4/30/24	Developed by: Air Care Team
Approved by: Dr. Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature: 	Signature:
Department Numbers: 3203	

I. **PURPOSE:** To provide supportive care to the patient, to minimize additional oxygen demands secondary to increased metabolism, and to limit the spread of any infectious process.

II. **PATHOPHYSIOLOGY:**

Sepsis can be bacterial, viral, fungal, parasitic, and mycobacterial in nature. The infectious stimulus may spread into the circulation resulting in the release of a cascade of inflammatory mediators, which can damage the vascular endothelium, leading to tissue hypoperfusion and eventual organ failure. The net result is that the patient will exhibit signs and symptoms of septic shock.

Patients develop systemic responses to the infection including fever or hypothermia, shaking, chills, hyperventilation, tachycardia, petechial or ecchymotic rash, confusion, agitation, lethargy, or coma. The child that develops septic shock may have delayed capillary refill, cyanosis, oliguria, or anuria, altered mental status, and hypoxemia. Several scoring systems exist (BOMBARD, SIRS, etc.) to identify and risk stratify sepsis. Monitoring tissue hypoperfusion (low MAP or elevated serum lactate levels) is the gold standard.

Meningitis is an inflammation of the meningeal coverings of the brain. The inflammation can be viral or bacterial. Nonbacterial (aseptic or viral) meningitis is usually associated with other diseases, such as measles, mumps, herpes, enteroviruses, & leukemia. Viral meningitis is non-purulent and usually resolves spontaneously. Bacterial meningitis is most commonly caused by one of the 3 following organisms: *Haemophilus influenzae*, *Neisseria meningitis*, and *Streptococcus pneumoniae*. Meningococcal meningitis is readily transmitted by droplet infection from nasopharyngeal secretions. Appropriate precautions and PPE must be utilized.

III. **DEFINITIONS:**

IV. **DEPARTMENT GUIDELINE:**

- A. Assess airway, breathing, circulation, and CNS by evaluation of the level of consciousness as per *GUID3203-PED001 – Pediatric General Management*.
- B. Assess the patient for rashes, petechial hemorrhages, ecchymosis, fever, nuchal rigidity, headache, and mental status changes. Physical examination in young infants should include assessment of anterior fontanel.
- C. Inhaled bronchodilators may be considered to optimize ventilation per *GUID3203-PED014 – Reactive Airway Disease* and *GUID3203-PR012 – Inhalation Therapy*.
- D. Fluid resuscitation

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- i. If signs of poor perfusion, administer 0.9% NS or Lactated Ringers 20 ml/kg IV, may be repeated per *GUID3203-PR015 – Intravenous Fluid Therapy*.
- ii. Most hypovolemic children will require at least 40-60 ml/kg in the first hour and those with sepsis may require up to 80 ml/kg in the first hour.
- iii. To help deliver fluid rapidly: Place an in-line 3-way stopcock in the IV tubing system. Deliver fluid by using a 30- to 60-mL syringe to push fluids through the stopcock or use a pressure bag (beware of risk of air embolism) or a rapid infusion device.
- iv. Assess IV patency and determine amount of volume previously received
- A. Antibiotics should be initiated when appropriate on inter-facility transfers prior to transport at the sending or receiving physician orders. Continue antiviral therapy if initiated by referring facility.
- B. Meningitis can cause hypoglycemia and hyponatremia.
 - v. Check blood glucose level. If < 60 d/L treat per *GUID3203-M021 – Hypoglycemia*.
 - vi. Seizures should be treated per *GUID3203- PED014 – Pediatric Seizures/Status Epilepticus*.
- C. Antipyretics should be considered on inter-facility transfers in febrile patients (**acetaminophen 15 mg/kg PO** or **ibuprofen 10 mg/kg PO** for children > 6 months of age).

V. DOCUMENTATION:

- A. EMS Charts

VI. REFERENCES:

- A. Pediatric Advanced Life Support Manual, 2020. American Heart Association.